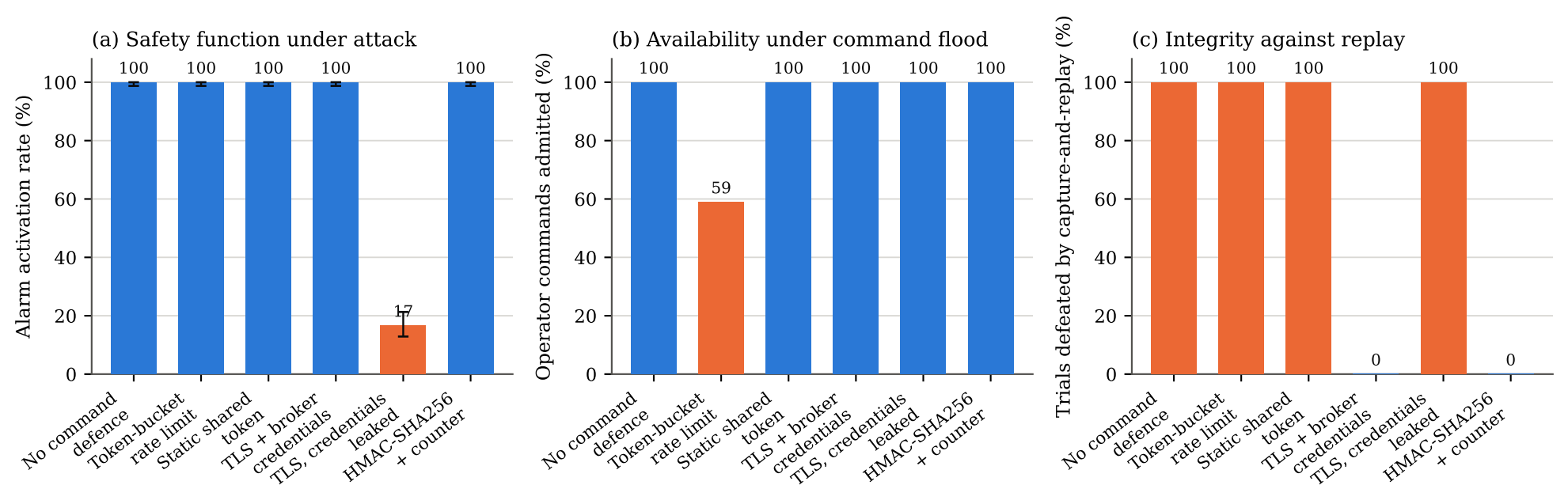
(a) Safety function under attack
Alarm activation rate (%)
0
20
40
60
80
100
100
100
100
100
17
100
No command defence
Token-bucket rate limit
Static shared token
TLS + broker credentials
TLS, credentials leaked
HMAC-SHA256 + counter
(b) Availability under command flood
Operator commands admitted (%)
100
59
100
100
100
100
(c) Integrity against replay
Trials defeated by capture-and-replay (%)
100
100
100
0
100
0